

Optimization I – Quiz 2

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1. Define the following concepts:

- a) Descent direction.
- b) Steepest descent direction.
- c) (Weak) Wolfe conditions.
- d) First Order Necessary Condition for Optimality.
- e) Second Order Necessary Condition for Optimality.
- f) Second Order Sufficient Condition for Optimality.
- g) Line search.

Proof.

a) A *descent direction* for f at $x \in \mathbb{R}^n$ is any vector $v \in \mathbb{R}^n$ such that

$$\nabla f(x)^\top v = v^\top \nabla f(x) < 0. \quad (1)$$

b) The *steepest descent direction* of f at x is given by

$$\operatorname{argmin}_{v \in \mathbb{R}^n} \langle \nabla f(x), v \rangle \quad (2)$$

c)

$$\begin{aligned} f(x_k + \alpha d_k) &\leq f(x_k) + \alpha \sigma_1 \nabla f(x_k)^\top d_k \\ \nabla f(x_k + \alpha d_k)^\top d_k &\geq \sigma_2 \nabla f(x_k)^\top d_k \end{aligned} \quad (3)$$

for $0 < \sigma_1 < \sigma_2 < 1$.

- d) If x^* is a local minimizer and f is continuously differentiable in an open neighborhood of x^* , then $\nabla f(x^*) = 0$.
- e) If x^* is a local minimizer of f and $\nabla^2 f$ exists and is continuous in an open neighborhood of x^* , then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite.
- f) Suppose that $\nabla^2 f$ exists and is continuous in an open neighborhood of x^* , and that $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite. Then x^* is a strict local minimizer of f .
- g) Minimization of a function in a descent direction. $f(x_k + \alpha_k d_k)$

□

2. Write the pseudocode for a line search method that satisfies the Armijo condition in Newton's direction, assuming that the Hessian is positive definite.

Algorithm 1: General Descent Method

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1 input  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ , starting point  $x_0 \in \mathbb{R}^n$ 
2 begin
3    $k := 1$ 
4   while converge criterion not satisfied
5     specify a descent direction  $d^k$  of  $f$  at  $x^k$ .
6     determine a step size  $\alpha_k > 0$  with  $f(x^k + \alpha_k d^k) < f(x^k)$ .
7     set  $x^{k+1} := x^k + \alpha_k d^k$ 
8     set  $k := k + 1$ 
9   end
10 end
    
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3. Consider the algorithm

$$x_{k+1} = x_k - \alpha_k H_k g_k \quad (4)$$

where $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, $f \in \mathcal{C}^2(\mathbb{R}^3)$, $g_k = \nabla f(x_k)$ and $H_k \in \mathbb{R}^{3 \times 3}$ is given by:

$$H_k = \begin{bmatrix} a^2 & -a & -a \\ -a & -a^2 & 3a \\ -a & 3a & -a^2 \end{bmatrix} \quad (5)$$

for some $a \in \mathbb{R}$. Suppose on the k -th iteration of the algorithm it holds that $g_k = [1, 1, 1]^\top$. Find the values of $a \in \mathbb{R}$ for which there exists $\alpha_k > 0$ that satisfies:

$$\alpha_k = \underset{\alpha > 0}{\operatorname{argmin}} f(x_k - \alpha H_k g_k) \quad (6)$$

Proof. We have

$$\begin{aligned}
 \alpha H_k g_k &= \alpha \begin{bmatrix} a^2 & -a & -a \\ -a & -a^2 & 3a \\ -a & 3a & -a^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\
 &= \alpha \begin{bmatrix} a^2 - 2a \\ -a^2 + 2a \\ -a^2 + 2a \end{bmatrix} \\
 &= \alpha a(a - 2) \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}
 \end{aligned} \quad (7)$$

We also have

$$\frac{d}{d\alpha} f(x_k - \alpha H_k g_k) = \nabla f(x_k - \alpha H_k g_k)^\top H_k g_k \quad (8)$$

At $\alpha = 0$,

$$\begin{aligned}
 \nabla f(x_k - \alpha H_k g_k)^\top H_k g_k &= g_k^\top H_k g_k \\
 &= -a(a - 2)
 \end{aligned} \quad (9)$$

If $\alpha > 0$, we need $-g_k^\top H_k g_k < 0$, so $0 < a < 2$.

□

4. Let $Q \in \mathbb{R}^{2 \times 2}$ be a symmetric matrix with eigenvalues $0 < \lambda_1 < \lambda_2$ and let u_1, u_2 be the eigenvectors associated with λ_1, λ_2 , respectively. Apply the steepest descent method with exact line search to minimize the function

$$f(x) = \frac{1}{2}x^\top Qx \quad (10)$$

starting from the point

$$x_0 = \frac{1}{\lambda_1}u_1 + \frac{1}{\lambda_2}u_2. \quad (11)$$

- a) Find the expression for x_k .
- b) Show that the algorithm converges to the optimal solution as $k \rightarrow \infty$
- c) Show that the rate $\frac{f(x_{k+1})}{f(x_k)}$ is strictly less than $\frac{\lambda_2 - \lambda_1}{\lambda_2 + \lambda_1}$